25901103

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026
Fourth SEMESTER
Discipline Specific Core Course (DSC) - MG4DSCBTS201 - CELL BIOLOGY
(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember (K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Fill in the blanks

Answer all questions

Each question carries **1** marks

1. The main function of chloroplast is to produce **carbohydrate/oxygen** through photosynthesis. [K] / [CO1]

2. **Active**.transport requires energy to move molecules against a concentration

gradient. [K] / [CO2]

3. Formation of shoot meristems in tissue culture is a **cytokinin** mediated response. [U] / [CO3]

4**. Lysosomes** are membrane bound organelles containing digestive enzymes. [K] / [CO1]

5. **Telomeres** are the protective caps at the end of chromosome. [K] / [CO2]

6. The region where spindle fibers attach is called the **centromere** [U] / [CO2]

7. The outermost layer of a plant cell is **cell wall** [K] / [CO1]

8. **Cholesterol** molecules regulate membrane fluidity. [K] / [CO2]

9. The three subunits of G-protein are **alpha (α), beta (β), and gamma (γ)** [K] / [CO3]

10. The process of generating energy in mitochondria is called **cellular respiration**.

[1x10 = 10]

Part B

ShortAnswer Questions

Answer any **10** questions

Each question carries **2** marks

11. Which cytoskeletal elements are associated with desmosomes? [K] / [CO2]
Intermediate filaments (specifically keratin filaments) are associated with desmosomes

12. Differentiate between karyokinesis and cytokinesis. [U] / [CO3]
Karyokinesis vs Cytokinesis
Karyokinesis: Division of the nucleus (mitosis or meiosis).
Cytokinesis: Division of the cytoplasm, resulting in separate daughter cells.

13. What is the role of intermediate filaments? [U] / [CO1]

Provide mechanical support and maintain cell shape.

14. What are selectins? Mention one of their functions. [U] / [CO2]
Mediate cell adhesion and rolling of leukocytes on endothelial cells.

15. What are stem cells? [K] / [CO3]
Undifferentiated cells that can differentiate into various cell types.

16. What is the role of histone proteins in nucleosomes? [K] / [CO2]
Package DNA into nucleosomes, regulating chromatin structure.

17. Who discovered cells and in which year? [K] / [CO1]
Discovered by: Robert Hooke, Year: 1665

18. What is meant by "critical mass" in quorum sensing? [U] / [CO3]
Minimum concentration of signaling molecules needed to trigger a response.

19. Comment on NOR [K] / [CO1]
NOR (Nucleolus Organizer Region) are region around which nucleolus forms, contains rRNA genes.

20. Name any two CDK inhibitors. [U] / [CO3]
p21, p27

21. Define prokaryotic cell and give one example. [K] / [CO1]
Cell lacking a true nucleus (e.g., E.coli)

22. Mention any two differences between apoptosis and necrosis. [K] / [CO3]
Apoptosis vs Necrosis
Apoptosis: Programmed cell death, regulated.
Necrosis: Uncontrolled cell death due to injury. **[2x10 = 20]**

Part C

Short Essay Questions
Answer any **5** questions
Each question carries **4** marks

23. Explain the different forms of chemical signaling in multicellular organisms. [U] / [CO3]
Chemical Signaling Forms

a. Endocrine: Hormones released into bloodstream (e.g., insulin).
b. Paracrine: Local signaling (e.g., neurotransmitters).
c. Autocrine: Cell signals itself.
d. Juxtacrine: Signaling via direct cell-cell contact.

24. Explain the role of ribosomes in protein synthesis. [K] / [CO1]
Ribosome**s** are small, non-membrane-bound structures found in all living cells. They are the sites of protein synthesis**.**
Translate mRNA into polypeptide chains (proteins).

25. Describe the functions of SER. [K] / [CO2]

The Smooth Endoplasmic Reticulum (SER) is a type of endoplasmic reticulum without ribosomes on its surface, so it appears smooth.
SER (Smooth Endoplasmic Reticulum) Functions
Lipid synthesis, detoxification, calcium storage

26. Describe the structure and formation of RER .Write two functions also. [K] / [CO2]
RER (Rough Endoplasmic Reticulum)
The Rough Endoplasmic Reticulum (RER) is a type of endoplasmic reticulum that has ribosomes attached to its surface**,** giving it a rough appearance under a microscope.
Formation: Involves membrane synthesis and ribosome attachment.
Functions:
1. Protein synthesis and folding.
2. Protein transport.

27. Describe the functions of vacuoles in plant cells. [K] / [CO1]
Storage, maintaining turgor pressure, waste disposal.

28. Explain the significance of crossing over during meiosis. [K] / [CO2]
Increases genetic diversity by exchanging genetic material between homologous chromosomes.

29. Describe the functions of golgi complex. [K] / [CO1]
Protein modification (glycosylation), sorting, packaging, and secretion (explain)

[4x5 = 20]